Taipei Veterans General Hospital

Department of Pathology and Laboratory Medicine
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## Sample Information

**Patient Name:** 王禎祺
**Gender:** Male
**ID No.:** F100576201
**History No.:** 45483554
**Age:** 75

**Ordering Doctor:** DOC3184H 江侑洵
**Ordering REQ.:** 0BVETNB
**Signing in Date:** 2022/05/13

**Path No.:** S111-99303
**MP No.:** F22046
**Assay:** Oncomine Focus Assay
**Sample Type:** FFPE
**Block No.:** S111-76210A
**Percentage of tumor cells:** 30%

**Reporting Doctor:** DOC5452C 周德盈 (Phone: 8#5452)

**Note:**

## Sample Cancer Type: Non-Small Cell Lung Cancer

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### Report Highlights

1 Relevant Biomarkers
1 Therapies Available
1 Clinical Trials

## Relevant Non-Small Cell Lung Cancer Variants

| Gene | Finding | Gene | Finding |
|---|---|---|---|
| ALK | None detected | NTRK1 | None detected |
| BRAF | None detected | NTRK2 | None detected |
| EGFR | None detected | NTRK3 | None detected |
| ERBB2 | None detected | RET | None detected |
| KRAS | ***KRAS p.(G12D) c.35G>A*** | ROS1 | None detected |
| MET | None detected | | |

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.04(003).The content of this report has not been evaluated or approved by FDA, EMA or other regulatory agencies.

## Relevant Biomarkers

| Tier | Genomic Alteration | Relevant Therapies (In this cancer type) | Relevant Therapies (In other cancer type) | Clinical Trials |
|---|---|---|---|---|
| IIC | *KRAS p.(G12D) c.35G>A*<br>KRAS proto-oncogene, GTPase<br>Allele Frequency: 52.73% | None | cabozantinib | 1 |

**Public data sources included in relevant therapies: FDA**[1], NCCN, **EMA**[2], ESMO
**Tier Reference:** Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

## Variant Details

### DNA Sequence Variants

| Gene | Amino Acid Change | Coding | Variant ID | Locus | Allele Frequency | Transcript | Variant Effect | Coverage |
|---|---|---|---|---|---|---|---|---|
| KRAS | p.(G12D) | c.35G>A | COSM521 | chr12:25398284 | 52.73% | NM_033360.4 | missense | 1995 |
| IDH1 | p.(G105=) | c.315C>T | . | chr2:209113192 | 55.45% | NM_005896.3 | synonymous | 2000 |
| FGFR4 | p.(P136L) | c.407C>T | . | chr5:176517797 | 99.75% | NM_213647.3 | missense | 2000 |
| EGFR | p.(Q787=) | c.2361G>A | . | chr7:55249063 | 99.80% | NM_005228.5 | synonymous | 2000 |
| MAP2K2 | p.(V64=) | c.192C>T | . | chr19:4117528 | 39.90% | NM_030662.4 | synonymous | 2000 |

## Biomarker Descriptions

**KRAS (KRAS proto-oncogene, GTPase)**

Background: The KRAS proto-oncogene encodes a GTPase that functions in signal transduction and is a member of the RAS superfamily which also includes NRAS and HRAS. RAS proteins mediate the transmission of growth signals from the cell surface to the nucleus via the PI3K/AKT/MTOR and RAS/RAF/MEK/ERK pathways, which regulate cell division, differentiation, and survival[1,2,3].

Alterations and prevalence: Recurrent mutations in RAS oncogenes cause constitutive activation and are found in 20-30% of cancers. KRAS mutations are observed in up to 10-20% of uterine cancer, 30-35% of lung adenocarcinoma and colorectal cancer, and about 60% of pancreatic cancer[4]. The majority of KRAS mutations consist of point mutations occurring at G12, G13, and Q61[4,5,6]. Mutations at A59, K117, and A146 have also been observed but are less frequent[7,8].

Potential relevance: The KRAS inhibitor, sotorasib[9], is approved (2021) for the treatment of adult patients with KRAS G12C-mutated locally advanced or metastatic non-small cell lung cancer (NSCLC). The FDA has granted breakthrough therapy designation (2021) to the small molecule inhibitor, adagrasib, for KRAS G12C positive in non-small cell lung cancer following prior systemic therapy[10].The small molecular inhibitor, RO-5126766, was also granted breakthrough designation (2021) alone for KRAS G12V mutant non-small cell lung cancer or in combination with defactinib, for KRAS mutant endometrial carcinoma and KRAS G12V mutant non-small cell lung cancer[11]. Additionally, onvansertib[12] was granted fast track designation (2020) for second-line treatment of patients with KRAS-mutated metastatic colorectal cancer (mCRC). The EGFR antagonists, cetuximab[13] and panitumumab[14], are contraindicated for treatment of colorectal cancer patients with KRAS mutations in exon 2 (codons 12 and 13), exon 3 (codons 59 and 61), and exon 4 (codons 117 and 146)[8]. Additionally, KRAS mutations are associated with poor prognosis in NSCLC[15].

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## Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types ✖ No evidence

### KRAS p.(G12D) c.35G>A

| Relevant Therapy | FDA | NCCN | EMA | ESMO | Clinical Trials* |
|---|---|---|---|---|---|
| cabozantinib | ✖ | ✖ | ✖ | ○ | ✖ |
| RMC-4630, pembrolizumab | ✖ | ✖ | ✖ | ✖ | ● (I/II) |

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

## Relevant Therapy Details

### Current ESMO Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

ESMO information is current as of 2022-02-28. For the most up-to-date information, search www.esmo.org.

### KRAS p.(G12D) c.35G>A

○ **cabozantinib**

**Cancer type:** Thyroid Gland Medullary Carcinoma **Variant class:** RAS mutation

**ESMO Level of Evidence/Grade of Recommendation:** II / C

**Population segment (Line of therapy):**
- Metastatic (First-line therapy)

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Thyroid Cancer [Annals of Oncology 30: 1856–1883, 2019 doi:10.1093/annonc/mdz400]

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## Clinical Trials in Taiwan region:

## Clinical Trials Summary

### KRAS p.(G12D) c.35G>A

| NCT ID | Title | Phase |
|---|---|---|
| NCT04418661 | A Phase 1/2, Open-label, Multicenter, Dose Escalation and Dose Expansion Study of SAR442720 in Combination With Pembrolizumab in Patients With Advanced Malignancies | I/II |

## Alerts Informed By Public Data Sources

### Current FDA Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

FDA information is current as of 2022-03-16. For the most up-to-date information, search www.fda.gov.

### KRAS p.(G12D) c.35G>A

#### cetuximab

**Cancer type:** Colorectal Cancer **Label as of:** 2021-09-24 **Variant class:** KRAS G12 mutation

**Indications and usage:**

Erbitux® is an epidermal growth factor receptor (EGFR) antagonist indicated for treatment of:

Head and Neck Cancer

- Locally or regionally advanced squamous cell carcinoma of the head and neck in combination with radiation therapy.
- Recurrent locoregional disease or metastatic squamous cell carcinoma of the head and neck in combination with platinum-based therapy with fluorouracil.
- Recurrent or metastatic squamous cell carcinoma of the head and neck progressing after platinum-based therapy.

Colorectal Cancer

K-Ras wild-type, EGFR-expressing, metastatic colorectal cancer as determined by FDA-approved test

- in combination with FOLFIRI for first-line treatment,
- in combination with irinotecan in patients who are refractory to irinotecan-based chemotherapy,
- as a single agent in patients who have failed oxaliplatin- and irinotecan-based chemotherapy or who are intolerant to irinotecan.

Limitations of Use: Erbitux® is not indicated for treatment of Ras-mutant colorectal cancer or when the results of the Ras mutation tests are unknown.

BRAF V600E Mutation-Positive Metastatic Colorectal Cancer (CRC)

- in combination with encorafenib, for the treatment of adult patients with metastatic colorectal cancer (CRC) with a BRAF V600E mutation, as detected by an FDA-approved test, after prior therapy.

**Reference:**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/125084s279lbl.pdf

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## KRAS p.(G12D) c.35G>A (continued)

### panitumumab

**Cancer type:** Colorectal Cancer **Label as of:** 2021-08-25 **Variant class:** KRAS G12 mutation

**Indications and usage:**

VECTIBIX® is an epidermal growth factor receptor (EGFR) antagonist indicated for the treatment of wild-type RAS (defined as wild-type in both KRAS and NRAS as determined by an FDA-approved test for this use) metastatic colorectal cancer (mCRC):

- In combination with FOLFOX for first-line treatment.
- As monotherapy following disease progression after prior treatment with fluoropyrimidine, oxaliplatin, and irinotecan-containing chemotherapy.
- Limitation of Use: VECTIBIX® is not indicated for the treatment of patients with RAS-mutant mCRC or for whom RAS mutation status is unknown.

**Reference:**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/125147s210lbl.pdf

### defactinib + RO-5126766

**Cancer type:** Endometrial Carcinoma **Variant class:** KRAS mutation

**Supporting Statement:**
The FDA has granted Breakthrough Designation to the small molecule inhibitor, RO-5126766 alone for KRAS G12V mutant non-small cell lung cancer or in combination with defactinib, for KRAS mutant endometrial carcinoma and KRAS G12V mutant non-small cell lung cancer.

**Reference:**

https://investor.verastem.com//news-releases/news-release-details/verastem-oncology-receives-breakthrough-therapy-designation-vs

### bevacizumab + onvansertib + FOLFIRI

**Cancer type:** Colorectal Cancer **Variant class:** KRAS mutation

**Supporting Statement:**
The FDA has granted Fast Track Designation to the Polo-like Kinase 1 (PLK1) inhibitor, onvansertib, in combination with FOLFIRI and bevacizumab, for KRAS mutations in metastatic colorectal cancer in the second line.

**Reference:**

https://cardiffoncology.investorroom.com/2020-05-28-Cardiff-Oncology-Announces-Fast-Track-Designation-Granted-by-the-FDA-to-Onvansertib-for-Second-Line-Treatment-of-KRAS-Mutated-Colorectal-Cancer

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## Current NCCN Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

NCCN information is current as of 2022-02-28. For the most up-to-date information, search www.nccn.org.
For NCCN International Adaptations & Translations, search www.nccn.org/global/international_adaptations.aspx.

## KRAS p.(G12D) c.35G>A

### cetuximab

**Cancer type:** Colon Cancer **Variant class:** KRAS exon 2 mutation

**Summary:**
NCCN Guidelines® include the following supporting statement(s):
- "Patients with any known KRAS mutation (exon 2, 3, 4) or NRAS mutation (exon 2, 3, 4) should not be treated with either cetuximab or panitumumab."

**Reference:** NCCN Guidelines® - NCCN-Colon Cancer [Version 1.2022]

### cetuximab

**Cancer type:** Rectal Cancer **Variant class:** KRAS exon 2 mutation

**Summary:**
NCCN Guidelines® include the following supporting statement(s):
- "Patients with any known KRAS mutation (exon 2, 3, 4) or NRAS mutation (exon 2, 3, 4) should not be treated with either cetuximab or panitumumab."

**Reference:** NCCN Guidelines® - NCCN-Rectal Cancer [Version 1.2022]

### panitumumab

**Cancer type:** Colon Cancer **Variant class:** KRAS exon 2 mutation

**Summary:**
NCCN Guidelines® include the following supporting statement(s):
- "Patients with any known KRAS mutation (exon 2, 3, 4) or NRAS mutation (exon 2, 3, 4) should not be treated with either cetuximab or panitumumab."

**Reference:** NCCN Guidelines® - NCCN-Colon Cancer [Version 1.2022]

### panitumumab

**Cancer type:** Rectal Cancer **Variant class:** KRAS exon 2 mutation

**Summary:**
NCCN Guidelines® include the following supporting statement(s):
- "Patients with any known KRAS mutation (exon 2, 3, 4) or NRAS mutation (exon 2, 3, 4) should not be treated with either cetuximab or panitumumab."

**Reference:** NCCN Guidelines® - NCCN-Rectal Cancer [Version 1.2022]

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## Current EMA Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

EMA information is current as of 2022-03-16. For the most up-to-date information, search www.ema.europa.eu/ema.

### KRAS p.(G12D) c.35G>A

#### cetuximab, cetuximab + oxaliplatin

**Cancer type:** Colorectal Cancer **Label as of:** 2020-01-30 **Variant class:** KRAS exon 2 mutation

**Reference:**

https://www.ema.europa.eu/en/documents/product-information/erbitux-epar-product-information_en.pdf

#### panitumumab + oxaliplatin

**Cancer type:** Colorectal Cancer **Label as of:** 2021-12-07 **Variant class:** KRAS exon 2 mutation

**Reference:**

https://www.ema.europa.eu/en/documents/product-information/vectibix-epar-product-information_en.pdf

## Current ESMO Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

ESMO information is current as of 2022-02-28. For the most up-to-date information, search www.esmo.org.

### KRAS p.(G12D) c.35G>A

#### cetuximab

**Cancer type:** Colorectal Cancer **Variant class:** KRAS exon 2 mutation

**Summary:**
ESMO Clinical Practice Guidelines include the following supporting statement:

- "It has been demonstrated that the (potential) benefit of anti-EGFR antibodies in all treatment lines and either as a single agent or in combination with any chemotherapy regimen is limited to patients in whom a RAS mutation is excluded. It was shown that the 'expanded RAS' analysis (also including the detection of mutations in exons 3 and 4 of the KRAS gene as well as mutations in the NRAS [exons 2-4] gene) is superior to the KRAS (exon 2) analysis in predicting both more efficacy in the expanded RAS wild-type (WT) patients and a potential detrimental effect in patients harbouring any RAS mutation in their tumour genome [II/A]."

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Colorectal Cancer [Ann Oncol (2014) 25 (suppl 3): iii1-iii9. (eUpdate: 20 September 2016; Corrigendum: 21 July 2015)]

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## KRAS p.(G12D) c.35G>A (continued)

### cetuximab + chemotherapy

**Cancer type:** Colorectal Cancer **Variant class:** KRAS exon 2 mutation

**Summary:**
ESMO Clinical Practice Guidelines include the following supporting statement:

- "It has been demonstrated that the (potential) benefit of anti-EGFR antibodies in all treatment lines and either as a single agent or in combination with any chemotherapy regimen is limited to patients in whom a RAS mutation is excluded. It was shown that the 'expanded RAS' analysis (also including the detection of mutations in exons 3 and 4 of the KRAS gene as well as mutations in the NRAS [exons 2-4] gene) is superior to the KRAS (exon 2) analysis in predicting both more efficacy in the expanded RAS wild-type (WT) patients and a potential detrimental effect in patients harbouring any RAS mutation in their tumour genome [II/A]."
- "Thus, the activity of the anti-EGFR antibodies is confined to RAS WT tumours (and not only KRAS WT tumours). This is true for the combinations of cetuximab or panitumumab alone or with irinotecan- and oxaliplatin-based regimens. Treatment with anti-EGFR antibodies may even harm patients with a RAS mutation, especially when combined with oxaliplatin [I/A]."

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Colorectal Cancer [Ann Oncol (2014) 25 (suppl 3): iii1-iii9. (eUpdate: 20 September 2016; Corrigendum: 21 July 2015)]

### panitumumab

**Cancer type:** Colorectal Cancer **Variant class:** KRAS exon 2 mutation

**Summary:**
ESMO Clinical Practice Guidelines include the following supporting statement:

- "It has been demonstrated that the (potential) benefit of anti-EGFR antibodies in all treatment lines and either as a single agent or in combination with any chemotherapy regimen is limited to patients in whom a RAS mutation is excluded. It was shown that the 'expanded RAS' analysis (also including the detection of mutations in exons 3 and 4 of the KRAS gene as well as mutations in the NRAS [exons 2-4] gene) is superior to the KRAS (exon 2) analysis in predicting both more efficacy in the expanded RAS wild-type (WT) patients and a potential detrimental effect in patients harbouring any RAS mutation in their tumour genome [II/A]."

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Colorectal Cancer [Ann Oncol (2014) 25 (suppl 3): iii1-iii9. (eUpdate: 20 September 2016; Corrigendum: 21 July 2015)]

### panitumumab + chemotherapy

**Cancer type:** Colorectal Cancer **Variant class:** KRAS exon 2 mutation

**Summary:**
ESMO Clinical Practice Guidelines include the following supporting statement:

- "It has been demonstrated that the (potential) benefit of anti-EGFR antibodies in all treatment lines and either as a single agent or in combination with any chemotherapy regimen is limited to patients in whom a RAS mutation is excluded. It was shown that the 'expanded RAS' analysis (also including the detection of mutations in exons 3 and 4 of the KRAS gene as well as mutations in the NRAS [exons 2-4] gene) is superior to the KRAS (exon 2) analysis in predicting both more efficacy in the expanded RAS wild-type (WT) patients and a potential detrimental effect in patients harbouring any RAS mutation in their tumour genome [II/A]."
- "Thus, the activity of the anti-EGFR antibodies is confined to RAS WT tumours (and not only KRAS WT tumours). This is true for the combinations of cetuximab or panitumumab alone or with irinotecan- and oxaliplatin-based regimens. Treatment with anti-EGFR antibodies may even harm patients with a RAS mutation, especially when combined with oxaliplatin [I/A]."

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Colorectal Cancer [Ann Oncol (2014) 25 (suppl 3): iii1-iii9. (eUpdate: 20 September 2016; Corrigendum: 21 July 2015)]

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## Signatures

**Testing Personnel:** Shin-Ying Lin 2022.05.16 15:03:59 +08'00'

**Laboratory Supervisor:** 2022.05.16 15:04:11 +08'00'

**Pathologist:** 2022.05.16 15:04:29 +08'00'

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## References


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